Highlights

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

Author names to be supplied

- Interpolation order, bounds and integral conservation require separate tests.
- Global correction preserves mass but does not recover a diffused profile.
- An older-history error perturbs the next step even when the latest is exact.
- Control-volume transfer retains mass with little Euler transition broadening.
- A load ranking remains unresolved despite better integral preservation.

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

Author names to be supplied[a]

[a]*Affiliation to be supplied, City, Country*

ARTICLE INFO

Keywords:
Solution transfer
Remeshing
Conservative interpolation
Galerkin projection
Backward differentiation
Finite-volume methods

ABSTRACT

Mesh replacement in an unsteady simulation changes both the spatial representation of the solution and the histories used by the time integrator. This study examines how these changes affect a second-order backward differentiation restart, comparing nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and a conservative control-volume transfer. Static tests separate approximation order, boundedness, and integral preservation. Under repeated Euler remeshing, globally corrected first-order interpolation preserves the cumulative density integral to roundoff while broadening the transition by 18–26 reference cells. The control-volume transfer retains the integral with at most 0.07 cell of broadening, comparable to second-order Galerkin projection in spatial fidelity. Controlled airfoil restarts show that an older-history discrepancy alone produces a finite next-step lift response even when the latest history is exact; an exact residual identity explains the signed contributions of the two histories. In Reynolds-averaged Navier–Stokes calculations with two turbulence closures, Galerkin projection reduces monitored turbulence-integral defects by factors of approximately 37–247 relative to second-order nearest-neighbor Taylor interpolation, while the control-volume transfer keeps every monitored defect below 5×10^{-12} at the stated processing stages. These transfer-level differences do not establish an aerodynamic ranking: paired load differences remain sensitive to time-step refinement, with incomplete inner convergence for one closure. The results show that integral preservation, spatial fidelity, and the response of transferred histories are distinct aspects of restart accuracy.

1. Introduction

Adaptive remeshing replaces the computational mesh during a simulation, and every replacement is a discrete restart: before time integration can continue, the numerical solution must be represented on the new mesh. Because the transfer is repeated at every adaptation, its errors can accumulate, and time-accurate adaptation frameworks therefore treat solution transfer together with the spatial and temporal discretization [1–3]. For a multistep time integrator, the restart involves more than the current state: every solution level used by the integrator must be supplied consistently on the receiving mesh. The question is then which properties a transfer operator needs in order to serve as such a restart.

The transfer literature characterizes an operator through three properties of a single field, each of which answers a different question. Interpolation order describes the approximation of smooth fields; boundedness, enforced by limiting, controls new extrema; and conservation constrains selected integrals or moments. Common-refinement and Galerkin approaches provide accurate conservative transfer between nonmatching meshes [4, 5], while constrained formulations preserve additional discrete properties, including norms and weak incompressibility [6]. Conservative remapping instead transfers integrals of a reconstructed source field to individual receiving cells [7–10]. Conservative Galerkin transfer also supports time-accurate anisotropic adaptation, including high-order implicit formulations with bounded transfer near discontinuities [3, 11]. The three properties are not interchangeable: preserving a domain integral leaves the spatial distribution undetermined, and enforcing bounds can alter the raw approximation or projection.

In a multistep restart, these single-field properties no longer describe the whole disturbance. The receiving discretization advances from transferred histories instead of states generated by its own previous steps, so an older-history error can perturb the next residual even when the latest history is exact. Limiting, representation changes, positivity enforcement, and boundary conditions can further modify the transferred field before the solver uses it. A latest-state interpolation norm or a conservation defect alone therefore leaves part of the restart disturbance unmeasured. Related work on balance-preserving transfer has shown that retaining a discrete equilibrium can prevent interpolation-induced disturbances in shallow-water calculations [12, 13]. Here the question concerns the histories of a compressible-flow time integrator and their effect on subsequent loads.

ORCID(s):

The study examines how spatial transfer properties and history discrepancies jointly affect a vertex-centered finite-volume calculation advanced with the second-order backward differentiation formula (BDF2). Four transfer families—nearest-neighbor Taylor interpolation (NN), barycentric interpolation (BAR), continuous Galerkin projection (GP), and conservative control-volume transfer (CV)—are compared. Numerical suffixes, as in NN1 or GP2, denote nominal smooth-field order. A bounded global correction (GC) isolates the effect of enforcing a domain integral. The CV construction uses receiving median-dual cells as test volumes, providing a comparator to Galerkin projection whose conservation constraints refer directly to a reconstruction compatible with the solver integral.

The contribution is a controlled separation of spatial fidelity, integral preservation, and multistep response. Smooth and discontinuous fields identify properties of a single transfer; repeated Euler remeshing reveals their cumulative consequences. Independent airfoil meshes then allow the latest and older BDF2 discrepancies to be varied separately. The standard BDF2 history weights give an exact residual-forcing identity, while the computed restarts establish how those contributions appear in the state and lift response. This connects the spatial comparisons to a temporal consequence that a single-field transfer norm does not describe.

Fixed-body Reynolds-averaged Navier–Stokes (RANS) calculations extend the comparison to turbulence transport and viscous loads. They distinguish measured differences in transfer-integral defects from aerodynamic differences that remain sensitive to the receiving calculation's numerical settings. All flow experiments use prescribed replacements on fixed domains and native discrete references: three Riemann deformation realizations per amplitude, two in each repeated Euler- and RANS-airfoil comparison, and one in the RANS sensitivity tests. The resulting evidence concerns restart behavior under these controlled replacements rather than the performance of a complete adaptive strategy.

2. Solution-transfer operators

Each transfer consists of a raw target-field construction and constraints on the state returned to the flow solver. The construction families are nearest-neighbor Taylor interpolation (NN), barycentric interpolation (BAR), continuous Galerkin projection (GP), and conservative control-volume transfer (CV). Numerical suffixes denote nominal smooth-field accuracy, not polynomial degree: BAR2 is second-order barycentric interpolation, whereas GP2 and CV2 use piecewise-linear fields. A bounded global correction (GC) adjusts the integral of an existing target field and is treated separately from these constructions.

2.1. Representation and derivative reconstruction

Source and target meshes, A and B, cover the same discretized domain Ω. The nodal state on mesh M is U_M, and its finite-volume mass matrix is $M_M = \mathrm{diag}(w_i^M) \otimes I$, where $w_i^M = |C_i^M|$ is the positive measure of the median-dual control volume C_i^M of vertex i, I is the identity on the state components, and $\otimes$ denotes the Kronecker product. For one scalar component ϕ, the solver integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \tag{1}$$

All methods return values at the flow-solver vertices. Mean-flow density, momentum, and total energy are transferred directly in conserved-variable form; E denotes specific total energy, so the energy inventory is ρE. Nodal interpolation is not control-volume averaging, even when Eq. (1) is used to assess its conservation.

Higher-order transfer requires derivatives recovered from neighboring values. A weighted least-squares fit uses

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \tfrac{1}{2} r_{ij}^T H_i r_{ij}, \qquad r_{ij} = x_j - x_i. \tag{2}$$

Here g_i and H_i approximate the gradient and Hessian of ϕ at source vertex i, and r_{ij} is the displacement to a neighboring vertex. A degree-one fit omits the Hessian. With smooth data, a full-rank scaled stencil, and controlled mesh geometry, the linear fit gives an $O(h)$ gradient error; the quadratic fit gives $O(h^2)$ gradient and $O(h)$ Hessian errors. Anisotropy, weighting, and stencil conditioning can degrade these estimates [14–17]. Third-order variants use the quadratic reconstruction, and their achieved order is verified numerically.

2.2. Nearest-neighbor and barycentric interpolation

Let $n(j)$ be the source vertex closest to target point x_j^B, and $r_j = x_j^B - x_{n(j)}^A$. Nearest-neighbor Taylor transfer gives

$$(\mathcal{I}_{AB}^{\mathrm{NN1}} \phi_A)_j = \phi_{A,n(j)}, \tag{3}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN2}}\phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \tag{4}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN3}}\phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \tfrac{1}{2} r_j^T H_{n(j)} r_j. \tag{5}$$

The variants respectively copy the donor value, include its local slope, and include its curvature. For $|r_j| = O(h)$ and sufficiently accurate derivatives, their smooth-field errors are $O(h)$, $O(h^2)$, and $O(h^3)$. The donor changes across source-node Voronoi boundaries, so the transferred value need not vary continuously with target position.

Barycentric interpolation instead uses the source simplex containing the target point. A conforming triangulation or tetrahedralization of the original mesh provides the search geometry, without adding control-volume vertices. In spatial dimension d, a simplex has vertices $x_{i_r}^A$, $r = 1, \ldots, d+1$. Its barycentric coordinates satisfy

$$x_j^B = \sum_{r=1}^{d+1} \lambda_r x_{i_r}^A, \qquad \sum_r \lambda_r = 1, \qquad \lambda_r \geq 0. \tag{6}$$

The same geometric weights define the second-order interpolant and its gradient-corrected extension:

$$(\mathcal{I}_{AB}^{\mathrm{BAR2}}\phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \tag{7}$$

$$(\mathcal{I}_{AB}^{\mathrm{BAR3}}\phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \tfrac{1}{2} g_{i_r} \cdot (x_j^B - x_{i_r}^A)\right]. \tag{8}$$

BAR2 reproduces affine fields and is bounded by the containing-simplex values. For a quadratic field with exact vertex gradients, the half-gradient term cancels its interpolation remainder; BAR3 is consequently third-order consistent. Shared values and gradients give continuity across conforming simplex interfaces. Neither method imposes a source-to-target integral constraint.

Derivative limiting changes these properties independently of nominal order. Multiplying the Taylor corrections by $0 \leq \Phi_i \leq 1$ reduces NN2 or NN3 to NN1 when $\Phi_i = 0$, but reduces BAR3 to BAR2. The smooth verification is unlimited; discontinuous tests also use Barth–Jespersen (BJ) limiting [18]. A limiter derived from linear increments does not automatically bound a quadratic reconstruction at every evaluation point. Extrema must therefore be checked for the actual returned field.

2.3. Galerkin projection and cross-mesh integration

Continuous Galerkin projection determines a coupled target field rather than evaluating target vertices independently. Let ϕ_A^h be the piecewise-polynomial source representation and V_B^p the continuous degree-p Lagrange space on the target simplicial mesh. The projected field $\phi_B^h \in V_B^p$ satisfies

$$(\phi_B^h - \phi_A^h, v_B)_{L_2(\Omega)} = 0 \quad \text{for every } v_B \in V_B^p. \tag{9}$$

This is the best L_2 approximation of the source representation ϕ_A^h in V_B^p [4, 5]. Writing $\phi_B^h = \sum_j v_j N_j^B$, where N_j^B are target nodal basis functions and v_j their coefficients, gives

$$H_B v = b, \qquad (H_B)_{ij} = \int_\Omega N_i^B N_j^B \,\mathrm{d}x, \qquad b_i = \int_\Omega N_i^B \phi_A^h \,\mathrm{d}x. \tag{10}$$

The sparse positive-definite matrix H_B couples basis functions sharing target elements. It differs from the diagonal finite-volume matrix M_B used for time integration.

A target element can cross several source elements. Although a target basis function has one polynomial expression there, the source field changes expression at each source-element boundary. The cross-mesh load must therefore be integrated as

$$b_i = \sum_{K_B} \sum_{K_A:\, |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h \,\mathrm{d}x. \tag{11}$$

Here K_A and K_B are source and target simplices, and $|K_A \cap K_B|$ is their intersection measure. On each intersection, both expressions are fixed polynomials. These intersections form the common refinement and permit polynomial-exact

quadrature: degree two for P^1/P^1 transfer and degree four for P^2/P^2 transfer on affine simplices [19, 20]. Complete, nonoverlapping coverage is required. The intersection geometry and coupled solve distinguish projection from the point searches used by NN and BAR.

GP2 uses P^1 source and target spaces. GP3 constructs a P^2 source representation, including internally reconstructed edge values, and solves for a complete quadratic target field. Only original-vertex values are then supplied to the flow solver; the edge construction and the consequence of this restriction are detailed in Appendix A. On mixed-element meshes the spaces are P^p on conforming simplicial subdivisions, not Q^p on the original elements. Nominal approximation orders are two and three under smoothness and mesh-stability assumptions; accuracy after vertex restriction is assessed separately.

2.4. Conservative control-volume transfer

Conservative control-volume transfer (CV) constrains the transferred field over the finite-volume cells of the receiving mesh instead of against continuous basis functions. Let C_i^M be the median-dual control volume of vertex i on mesh M, so that $w_i^M = |C_i^M|$, and let $C_i^M \cap K$ be its piece in element K. Both meshes carry a continuous field $\phi_M^h = \sum_k \phi_{M,k} \psi_k^M$. On triangles and tetrahedra, $\psi_k^M = N_k^M$ is the P^1 hat function. On quadrilaterals, the subdivision joins each edge to the vertex-average center, whose field value is a weighted combination of the original nodal values. Analogous centered subdivisions are used on nonsimplicial three-dimensional elements. The dependent values are chosen to satisfy

$$\int_K \psi_k^M \, \mathrm{d}x = |C_k^M \cap K| \qquad \text{for every element } K \text{ and vertex } k \text{ of } M. \tag{12}$$

For a quadrilateral, the center weight of vertex k is $|C_k^M \cap K|/|K|$. Summation over elements gives $\int_\Omega \psi_k^M \, \mathrm{d}x = w_k^M$, and hence $\int_\Omega \phi_M^h \, \mathrm{d}x = Q_M(\phi_M)$: the field integral equals the solver integral, Eq. (1), on every supported element type. The source field is thus the continuous interpolant of the nodal solver values rather than a piecewise-constant or piecewise-linear reconstruction on the source control volumes. It reproduces the source integral Q_A, but not the individual contents $w_i^A \phi_{A,i}$ of the source control volumes: before limiting and boundary processing, CV transfer preserves the domain integral and the field integral over each receiving control volume, not the content of each source control volume. On parallelograms the weights reduce to the arithmetic vertex average, and the representation reproduces affine fields. On quadrilaterals that remain distorted it does not, because Eq. (12) is imposed in preference to affine reproduction.

The n_B target vertex values are defined by

$$\int_{C_j^B} \left(\phi_B^h - \phi_A^h \right) \, \mathrm{d}x = 0, \qquad j = 1, \ldots, n_B. \tag{13}$$

This is a Petrov–Galerkin condition with trial functions ψ_k^B and the indicator functions of the receiving control volumes as test functions. Intersection-based P^1-conservative interpolation [9, 10] instead preserves the integral over each receiving element through an element-wise linear reconstruction, limits its gradient, and recovers vertex values by explicit volume-weighted averaging over the elements incident to each vertex. Equation (13) imposes the integrals over the solver control volumes directly and therefore requires the solution of a sparse system. In matrix form,

$$\widetilde{H}_B \phi_B = \tilde{b}, \qquad (\widetilde{H}_B)_{jk} = \int_{C_j^B} \psi_k^B \, \mathrm{d}x, \qquad \tilde{b}_j = \int_{C_j^B} \phi_A^h \, \mathrm{d}x. \tag{14}$$

Because the functions ψ_k^B sum to one, the row sums of $\widetilde{H}_B$ are w_j^B; by Eq. (12), its column sums are w_k^B. The matrix has nonnegative entries and the element sparsity of H_B. It is symmetric on meshes of triangles and strictly convex quadrilaterals and on tetrahedral meshes: on a simplex, the vertex permutations exchange the median-dual pieces, and on a quadrilateral, the center weight of each vertex is proportional to its median-dual piece. Its entries depend only on target geometry and are evaluated in closed form.

The cross-mesh load is the integral of the piecewise-linear source representation over each receiving control volume. It is evaluated on intersections of the source subelements and target median-dual pieces,

$$\tilde{b}_j = \sum_{K_B \ni j} \sum_{K_A \colon |K_A \cap K_B| > 0} \int_{K_A \cap K_B \cap C_j^B} \phi_A^h \, \mathrm{d}x, \tag{15}$$

where K_B ranges over target elements incident to j and K_A over source subelements. Both meshes cover the same polygonal domain in the reported calculations; uncovered measure is below 10^{-20} of the domain. The target coefficients are obtained from Eq. (14) at a relative residual tolerance of 10^{-15}. They are nodal values, not the control-volume averages $\tilde{b}_j/w_j^B$.

Summing Eq. (13) over the receiving control volumes, which tile Ω, and using the column sums of $\widetilde{H}_B$ gives

$$Q_B(\phi_B) = \mathbf{1}^T \widetilde{H}_B \phi_B = \mathbf{1}^T \tilde{b} = \int_\Omega \phi_A^h \, \mathrm{d}x = Q_A(\phi_A), \tag{16}$$

to within integration and algebraic tolerances. The local statement, Eq. (13), concerns the field: $\int_{C_j^B} \phi_B^h \, \mathrm{d}x$ equals the source content of C_j^B. The nodal content $w_j^B \phi_{B,j}$ represented by M_B differs from it by $(\widetilde{H}_B \phi_B)_j - w_j^B \phi_{B,j}$, which sums to zero over the mesh but does not vanish individually. For $B = A$, $\tilde{b} = \widetilde{H}_A \phi_A$, and the transfer returns ϕ_A. If both representations reproduce affine fields, uniqueness of the solution of Eq. (14) makes the transfer exact for them. CV2 is consequently nominally second order on simplicial meshes and on quadrilaterals that remain parallelograms under refinement; persistent quadrilateral distortion reduces the attainable order. No higher-order CV variant is used.

On simplicial meshes, CV2 and GP2 seek the target field in the same space V_B^1, from the same P^1 source field and common refinement, and both solve a sparse system with the element coupling of H_B. They differ in the test space and therefore in the integrals they preserve. GP2 tests against the hat functions: H_B is the symmetric consistent mass matrix, the result is the L_2 best approximation, and the integral of the field and its coordinate moments are preserved over Ω, but not over individual control volumes. CV2 tests against the control-volume indicators: before limiting and away from the boundary trace, the field integral over every receiving control volume is preserved, and Eq. (16) holds for the solver integral, but the result is not a best approximation, and coordinate moments are not preserved in general. CV2 is thus a coupled transfer with local conservation constraints, not an explicit cell-by-cell remap: $\widetilde{H}_B^{-1}$, like H_B^{-1}, couples all target vertices. On meshes containing quadrilaterals, GP2 uses a conforming triangulation without added vertices, whereas CV2 uses the centered subdivision above with dependent center values.

The recovery matrix is symmetric positive definite on triangular, strictly convex quadrilateral, and tetrahedral meshes. Its symmetry follows from the median-dual construction; strict diagonal dominance gives positive definiteness on these element types. These properties do not extend generally to mixed three-dimensional cells. The numerical comparisons reported here are two-dimensional, and the principal accuracy limitation is the loss of affine reproduction on persistently distorted quadrilaterals.

CV2 denotes the limited transfer and CV2-u the unlimited construction. The flow comparisons use the same soft-bound tolerance as GP2, $\tau = 2 \times 10^{-3}$. Limiting and the subsequent boundary and admissibility treatment change the raw field, so the conservation statements must be distinguished from properties of the returned solver state.

2.5. Integral preservation, bounds, and the returned state

When the unconstrained target space contains constants, Eq. (9) preserves the integral of the complete finite-element field. It also preserves consistent coordinate moments represented by the test space. These identities do not imply conservation over arbitrary finite-volume cells, whose indicator functions generally lie outside a continuous finite-element space. CV2 uses these indicators as test functions; Eq. (13) preserves the field integral over each receiving control volume, and Eq. (16) the solver integral. Neither the Galerkin mass-matrix inverse nor $\widetilde{H}_B^{-1}$ is a local operator: unchanged geometry in a subregion does not guarantee unchanged transferred values there.

For GP2, consistent and solver-lumped integrals coincide when $w_i^M = \int_\Omega N_i^M \, \mathrm{d}x$ on both meshes. This condition holds on simplicial meshes; on the conforming triangulation of a quadrilateral, the hat-function integrals generally differ from the median-dual measures. The CV2 representation satisfies the corresponding condition, Eq. (12), on every supported element type. For GP3, source reconstruction and target vertex restriction introduce additional representation differences. An exact quadratic projection can acquire a second-order lumped-integral error after vertex restriction (Appendix A). A third-order pointwise approximation therefore does not imply a third-order conservation defect in the solver representation.

The bounded global correction can enforce Eq. (1) for any base transfer. For a raw target field ϕ_B^r, the correction c satisfies

$$\sum_j w_j^B c_j = Q_A(\phi_A) - Q_B(\phi_B^r), \qquad \ell_j \le \phi_{B,j}^r + c_j \le u_j. \tag{17}$$

The nodal lower and upper bounds are ℓ_j and u_j. The defect is redistributed among nodes with available capacity, with saturated nodes fixed [21, 22]. The requested integral must lie within the weighted bounds. A small integral defect does not bound each correction, and bounds imposed separately on conserved components do not ensure positive pressure.

GP and CV limiting constrain the target field using extrema of the overlapping source vertices. Hard limiting uses these bounds directly; soft limiting widens them by τ times the source range. Clipping is followed by redistribution that retains a selected global integral: the simplicial lumped total for GP and Q_B for CV. Once limiting is active, Galerkin orthogonality and best approximation need not survive. For CV, global redistribution likewise changes the individual receiving-volume integrals while retaining their sum.

Boundary values are taken from the source trace on the matching physical boundary. This replaces the unconstrained transfer there and can alter the local projection or control-volume identities. Where source and target wall vertices coincide, the trace retains the source wall condition; a change in the discretized surface can instead introduce a mismatch with the receiving wall orientation. The reported flow tests retain the boundary vertices, avoiding this geometric contribution.

For GP, CV, and globally corrected transfers, boundary changes and scalar floors are accompanied by integral-preserving redistribution. Remaining inadmissible states are conditioned while retaining the affected patch totals. These operations preserve the selected global integral but generally change local constraints. The distinction matters when interpreting conservation diagnostics: a property of the raw construction cannot automatically be assigned to the state subsequently used by the solver.

3. Multistep response to a mesh change

A mesh change perturbs a multistep calculation through each history entering the next physical step. Using the native target-mesh trajectory as a common reference separates this temporal effect from a change in the receiving evolution operator. The BDF2 history weights determine the residual disturbance exactly; the resulting state and load responses also depend on the receiving dynamics.

3.1. Native and transferred histories

Between mesh replacements the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \tag{18}$$

Here t is physical time and F_M is the assembled spatial residual, including fluxes, source terms, and the imposed boundary treatment. The state includes every transported variable required by the time integrator, including turbulence variables for RANS. Boundary constraints must be imposed consistently on the histories and on the equations advanced in time. At a constant time step, the target BDF2 equation is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t}\left(\tfrac{3}{2}Y - 2V + \tfrac{1}{2}W\right) + F_B(Y, t^{n+1}) = 0. \tag{19}$$

Here V and W are the latest and older histories. The mesh is stationary during each time step. A continuously moving mesh would require compatible geometric histories and a discrete geometric conservation law [23, 24]; Eq. (19) is not intended as that general formulation.

Let U_A^k and U_B^k denote native trajectories for the same physical problem on two meshes. The transferred histories and their target-referenced discrepancies are

$$\widehat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \qquad \varepsilon^k = \widehat{U}_B^k - U_B^k, \qquad k = n, n-1. \tag{20}$$

The effective map $\mathcal{I}_{AB}$ includes the processing needed to form admissible histories. The discrepancy ε^k combines interpolation error with differences between the native source and target discretizations. The native target state U_B^k supplies a common receiving evolution operator, not an exact continuum reference; Appendix A.1 gives the nonlinear decomposition.

3.2. Exact history forcing and conditional state response

At a common trial state, the transferred and native BDF2 equations differ only through their histories, and this difference is exact.

Proposition 1 (History forcing of transferred histories). *Let Φ_B be the constant-step BDF2 residual of Eq. (19), and let the transferred histories differ from the native target histories by ε^n and ε^{n-1}, Eq. (20). Then, for every trial state Y,*

$$\begin{aligned}\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) \\ = -\frac{M_B}{\Delta t}\eta^n, \qquad \eta^n = 2\varepsilon^n - \tfrac{1}{2}\varepsilon^{n-1}.\end{aligned} \tag{21}$$

PROOF. At the common trial state and time, the spatial residual and the term $3M_BY/(2\Delta t)$ cancel. The remaining difference is $(M_B/\Delta t)(-2\varepsilon^n + \varepsilon^{n-1}/2)$. □

The identity follows from the linear dependence of BDF2 on its histories, even when F_B is nonlinear and the discrepancies are finite. Boundary-modified residual rows require the corresponding boundary operation on the predicted forcing. This is distinct from a linear approximation to the solved-state response.

The solved-state difference $\delta^{n+1} = \widehat{U}_B^{n+1} - U_B^{n+1}$ obeys a different relation. Let r and $\hat{r}$ be the algebraic residuals remaining after the native and perturbed physical steps. Subtraction and expansion about the native target state give

$$\begin{aligned}A_B\delta^{n+1} &= M_B\eta^n + \Delta t(\hat{r} - r) - \Delta t\,\mathcal{N}_B(\delta^{n+1}) \\ A_B &= \tfrac{3}{2}M_B + \Delta t J_B, \qquad J_B = \frac{\partial F_B}{\partial U}.\end{aligned} \tag{22}$$

The Jacobian J_B is evaluated at the native target solution at t^{n+1}, and $\mathcal{N}_B$ is the nonlinear Taylor remainder. A locally Lipschitz Jacobian gives $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$. If the step is locally invertible, the perturbation is small, and the algebraic errors are negligible, then

$$\delta^{n+1} = R_B\eta^n + O(\Delta t\|\eta^n\|^2), \qquad R_B = A_B^{-1}M_B. \tag{23}$$

The bound is local to the mesh, state, time step, and norm. Limiter switching and incomplete inner convergence can invalidate this smooth-response approximation without invalidating Eq. (21).

A superscript $\star$ denotes a computed steady reference state. Table 1 assigns the same spatial discrepancy $\varepsilon^\star = \mathcal{I}_{AB}U_A^\star - U_B^\star$ to the two histories independently. Writing the corresponding responses as δ_f, δ_n, and δ_{n-1},

Table 1
History perturbations used to separate the BDF2 contributions. The spatial discrepancy $\varepsilon^\star$ is the same within each method; the native case supplies the receiving-mesh reference.

Configuration	ε^n	ε^{n-1}	η^n
Native	0	0	0
Both histories	$\varepsilon^\star$	$\varepsilon^\star$	$3\varepsilon^\star/2$
Latest only	$\varepsilon^\star$	0	$2\varepsilon^\star$
Older only	0	$\varepsilon^\star$	$-\varepsilon^\star/2$

local linearity predicts

$$\delta_n \simeq \tfrac{4}{3}\delta_f, \qquad \delta_{n-1} \simeq -\tfrac{1}{3}\delta_f, \qquad \delta_f \simeq \delta_n + \delta_{n-1}. \tag{24}$$

The older-only case isolates a disturbance invisible to a latest-state diagnostic. Because the imposed discrepancies are collinear, the predicted ratios describe directional response rather than the action of R_B on arbitrary fields.

3.3. Aerodynamic response

For a differentiable target-mesh output G_B, define $\Delta G^{n+1} = G_B(\widehat{U}_B^{n+1}) - G_B(U_B^{n+1})$. With g_B^{n+1} the gradient of G_B with respect to the state at the native target solution, Eq. (23) yields

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B\eta^n + O(\|\eta^n\|^2). \tag{25}$$

The load response depends on the spatial discrepancy weighted by the target dynamics and output sensitivity, not just its integral. In particular, preserving the source integral imposes, componentwise,

$$\mathbf{1}^T M_B \varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k). \tag{26}$$

The right-hand side need not vanish because the native discrete states can have different integrals. Global correction therefore need not reduce the target-referenced state or load discrepancy.

After the first step, the new solution difference and remaining older history affect subsequent evolution, so the initial load difference need not bound the later excursion. Disturbances away from the airfoil can produce delayed responses on acoustic or convective scales. Because neither the Galerkin mass-matrix inverse nor an implicit step has compact spatial support, the experiments identify arrival through finite-amplitude thresholds.

4. Numerical experiments

4.1. Static transfer verification

Static transfer tests exclude flow-solver evolution so that operator properties can be measured directly. The smooth asymmetric field is

$$\phi = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y)\right]. \tag{27}$$

Here ρ_∞ is the reference density, and L_x and L_y are the coordinate length scales. Structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes have nominal resolutions 64, 128, and 256; source resolution is half the target resolution. The relative nodal density error is $E_2 = [\sum_j (\rho_{B,j} - \rho(x_j^B))^2 / \sum_j \rho(x_j^B)^2]^{1/2}$, with equal vertex weights; it measures point-value accuracy rather than a volume-weighted field error. The effective spacing $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$ defines refinement ratios on the fixed domain. Pairwise orders are

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}. \tag{28}$$

Here E_k is the relative nodal error and h_k the effective spacing at refinement level k. A least-squares fit of $\log E$ against $\log h$ summarizes each three-level sequence. The discontinuous field has density 1.2 outside and 3.6 inside a circular inclusion of radius $0.25 \min(L_x, L_y)$, whose interface crosses the mesh. The discontinuous state has $T = 273(2 + \sigma)$ K, $\rho = 1.2(2 + \sigma)$, and $(u, v) = (100, 50)\sigma$ m/s, where $\sigma = 1$ inside the inclusion and -1 outside. Momentum and total energy follow from these primitive fields with $R = 287.05$ and $\gamma = 1.4$; the conserved fields are transferred independently. Positive density and internal energy determine admissibility. Unlimited and limited transfers are compared through density overshoot, undershoot, and the source-to-target lumped-integral defect.

4.2. Repeated Euler remeshing

The repeated-remeshing calculation asks whether the one-transfer properties measured above predict cumulative spatial fidelity. It uses a two-dimensional Euler calculation on $[0, 5] \times [0, 1]$. The uniform reference mesh has 501×101 vertices and spacing $h = 0.01$. The gas is ideal, with heat-capacity ratio $\gamma = 1.4$ and unit gas constant. Density, pressure, and temperature are denoted by ρ, p, and T, and Cartesian velocity components by u and v. Initially,

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1), & x < 2, \\ (0.125, 0, 0, 0.1), & x \geq 2, \end{cases} \tag{29}$$

so the left and right temperatures are 1 and 0.8. The upper and lower boundaries are inviscid slip walls. At $x = 0$ and $x = 5$, characteristic boundary conditions prescribe total pressure and temperature $(p_0, T_0) = (1, 1)$ and $(0.1, 0.8)$, respectively, with direction $(1, 0)$ at both ends.

The calculation advances 400 physical steps with $\Delta t = 10^{-3}$. Interior vertices receive fresh random displacements in x relative to the original mesh; boundaries remain fixed, and geometric distortion does not accumulate between successive meshes. For vertex i, let $a_{i,\min}$ be the smallest altitude of the incident corner triangles. The admissible streamwise displacement is bounded by $a_{i,\min}/s$, where the dimensionless deformation parameter s controls mesh perturbation amplitude. Values $s = 4$ and $s = 2$ are used; $s = 2$ permits twice the maximum displacement of $s = 4$.

Mesh replacement intervals of ten and two physical steps compare low- and high-frequency transfer. Both BDF2 histories are transferred at each mesh change. The low-frequency comparison contains 80 transfer evaluations; the high-frequency integral record contains 398. The fixed-mesh numerical reference uses the same initial and boundary conditions; differences from it include transfer error and the change in spatial discretization.

Density profiles are averaged in y with a common sampling procedure. The transition position is its 50% crossing, and its width is the distance between the 10% and 90% crossings. The differences Δx and Δw denote the transferred-calculation position and width minus their fixed-mesh reference values, expressed in cells of the $h = 0.01$ reference mesh. They describe a resolved density transition, without presuming a separately identified sharp shock. Density and energy integral changes are evaluated immediately across each transfer, separately from physical evolution. With $I_\rho = Q_M(\rho_M)$, each signed increment ΔI_ρ is the target integral minus the source integral; $\sum \Delta I_\rho$ accumulates these signed increments. NN1, NN1+GC, BAR2, GP2, and CV2 are compared at both amplitudes; additional methods extend the frequency comparison at $s = 4$.

4.3. Euler airfoil and controlled BDF2 histories

The airfoil calculations test consequences that cannot be isolated in the repeated Riemann problem: the independent contribution of the two BDF2 histories and the evolution of a transfer disturbance into an aerodynamic output. Euler flow over the National Advisory Committee for Aeronautics (NACA) 0012 airfoil is calculated at freestream Mach number $M_\infty = 0.1$ and angle of attack $\alpha = 6°$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4}$, $U_\infty = 0.118322$, and $\Delta t = 10^{-3}$. Here the subscript ∞ denotes freestream quantities, a_∞ is the sound speed, and U_∞ is the speed. The airfoil chord is denoted by c. The source and independent target meshes contain 16,255 and 22,508 vertices, respectively, with 32,077 and 44,583 triangles. Both retain the same 370 airfoil vertices and 63-vertex farfield polygon at $100c$, so the discretized surface is unchanged. Forces are pressure-only, with unit reference chord and area and a moment reference at $x/c = 0.25$; lift and drag coefficients are denoted C_L and C_D. Nonzero steady drag is treated as a numerical load, not skin friction or shock-induced drag.

Each native steady solution is continued with equal initial BDF2 histories. The history combinations in Table 1 are then formed on the target mesh and compared with its native continuation at the same physical step. State differences use the nondimensional volume-weighted norm $\|v\|_{M_B} = (v^T M_B v)^{1/2}$ over the stored mean-flow components. Supplementary tests also report the unweighted Euclidean norm $\|v\|_2$; these norms weight the near-body and large farfield cells differently. Relative superposition defects are

$$e^U_{\mathrm{sup}} = \frac{\|\delta_f - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_f\|_{M_B}}, \qquad e^{C_L}_{\mathrm{sup}} = \frac{|\Delta C_{L,f} - \Delta C_{L,n} - \Delta C_{L,n-1}|}{|\Delta C_{L,f}|}. \tag{30}$$

State-norm ratios are nonnegative; force ratios retain their sign.

The solver reports $-\Phi_B$. Its initial residuals are compared before an update using two common-trial pairs: older-only minus native at $U^\star_B$, and latest-only minus both-histories at $\widehat{U}^\star_B$. Their differences in η are $-\epsilon^\star/2$ and $+\epsilon^\star/2$, respectively. Double-precision residuals and a wall-normal/tangential split separate temporal forcing from slip-wall enforcement. Additional cancellation and amplitude tests are described in Appendix C.

A repeated-remeshing experiment perturbs vertices at wall distances between $0.01c$ and $20c$, while retaining the boundaries. NN2, GP2, and CV2 use matching deformation realizations, with 39 mesh replacements followed by evolution in a 400-step interval. Load differences are formed against the corresponding fixed-mesh trajectory. Slopes per 100 physical steps describe this finite window; they estimate neither long-time drift nor independent-sample uncertainty.

A separate single-transfer experiment perturbs narrow bands beginning at wall distances $0.05c$, $0.10c$, and $0.20c$. It uses GP2; CV2 repeats the longer $0.20c$-band calculation. Fixed pressure shells measure the root-mean-square (RMS) pressure difference from a common fixed-mesh reference, and force onset is defined by a fraction of the response maximum. For a shell at r inward of a band at d, the low-Mach acoustic scale is $(d - r)/(a_\infty \Delta t) \simeq 845(d - r)$ steps. Immediate disturbances extend up to approximately $0.08c$ inward of the displaced band; these shells are excluded from the threshold-dependent propagation fit.

4.4. Fixed-body RANS and numerical sensitivity assessment

The fixed-body RANS experiment examines whether the transfer differences identified in the preceding tests persist under turbulence transport and viscous loads. Repeated remeshing perturbs a nominally steady viscous state; the resulting between-method load differences are compared with the numerical sensitivity of the receiving calculation.

It retains $M_\infty = 0.1$ and $\alpha = 6°$, with chord Reynolds number $Re_c = 6 \times 10^6$ and the Spalart–Allmaras (SA) and shear-stress-transport (SST) closures [25, 26]. The coarse mesh has 23,271 vertices, 9,686 quadrilaterals in the wall layer and wake region, 26,851 triangles, and 256 airfoil vertices. The medium and fine meshes contain 32,102 and 48,228 vertices. All three retain the same first off-wall spacing. The farfield is at $100c$; the first off-wall spacing is approximately $5 \times 10^{-6}c$, and the quadrilateral region extends to about $0.11c$.

The calculations use the Stanford University Unstructured (SU2) solver [27]. Mean-flow convection uses the Harten–Lax–van Leer contact (HLLC) flux with a monotonic upstream-centered scheme for conservation laws (MUSCL) reconstruction; turbulence transport uses scalar upwinding without MUSCL reconstruction. BDF2 physical steps are solved by dual-time iteration at a Courant–Friedrichs–Lewy (CFL) number of 6. The nominal inner-iteration limit is 800 iterations. After ten iterations, the stopping test checks $\log_{10} \mathrm{rms}(r_\rho) \le -12$, where r_ρ is the nodal density-equation residual in the solver normalization. The dimensional physical step $\Delta t_0 = 10^{-3}$ s corresponds to the nondimensional value $\Delta t_0^* = 0.347224$. A superscript $*$ denotes quantities normalized with the solver reference scales. With $U_\infty^* = 0.1$, the 400-step interval spans approximately 13.9 chord flow-through times, where convective time is measured by tU_∞/c. A first-order startup step is ten times smaller. Each closure has a no-remeshing reference continued for 800 steps from matching histories.

On the coarse mesh, the steady SST solution reaches the density-residual target, whereas the SA solution reaches its 20,000-iteration limit at -11.2, with lift and drag stationary to 10^{-7} over the final 2,000 iterations. The steady SST solution on the medium mesh stops at approximately -11.1. These residual levels and the reference continuations are distinguished from convergence of the remeshed physical steps.

The same prescribed mesh sequences are used for NN2, limited GP2, and limited CV2. At each replacement, 12,602 vertices may move and 10,669 are fixed. No vertex with solver wall distance below $0.01c$ moves; the geometric mask uses distance to the nearest airfoil vertex, and part of the outer quadrilateral layer can move. The nominal RANS deformation uses $s = 4$, and a minimum corner-area ratio of 0.2 preserves valid cells. The boundaries remain fixed. Mesh replacement occurs every ten physical steps, independently of the evolving solution.

Both mean-flow and turbulence histories are transferred at the required time levels. SA transports the working variable $\tilde{\nu}$. For SST, k and ω denote turbulent kinetic energy and specific dissipation rate; their density-weighted values are transferred before primitive-variable positivity floors are applied. Wall distance and eddy viscosity are recomputed. For a transferred scalar ϕ, the source-relative defect is $[Q_B(\phi_B) - Q_A(\phi_A)]/Q_A(\phi_A)$; all monitored source integrals are nonzero in these cases. Turbulence inventories change under their transport equations and are not physical invariants. The SST integral diagnostic precedes the final positivity floor, so it characterizes transfer at that stage rather than conservation of the fully processed restart.

The pressure and viscous contributions to total drag are denoted by $C_{D,p}$ and $C_{D,v}$, respectively. For an aerodynamic coefficient C_Q and transfer method m, define the remeshing-induced response and paired method difference by

$$D_{m,Q}(t) = C_{Q,m}(t) - C_{Q,\mathrm{ref}}(t), \qquad P_Q(t) = D_{\mathrm{NN2},Q}(t) - D_{\mathrm{GP2},Q}(t). \tag{31}$$

Pressure and viscous contributions use identical surface quadrature for the remeshed and reference calculations. Over N common physical sample times t_i, the RMS of a scalar history q is $\mathcal{R}(q) = (N^{-1} \sum_i q(t_i)^2)^{1/2}$. For a modified numerical setting c, relative to the nominal setting denoted by subscript 0, sensitivity is measured by

$$S(q; c) = \frac{\mathcal{R}(q_c - q_0)}{\mathcal{R}(q_0)}. \tag{32}$$

This measures the difference between histories, not the fractional change of their RMS amplitudes. The latter is $[\mathcal{R}(q_c) - \mathcal{R}(q_0)]/\mathcal{R}(q_0)$.

The sensitivity assessment uses a common 38-sample window containing three mesh replacements. Inner-iteration limits are increased to 1600 for both closures and to 6400 for SST. Time-step refinement uses $\Delta t_0/2$ for both closures and $\Delta t_0/4$ for SA, with identical displacement fields applied at matching physical times and a separate no-remeshing reference for each step size. The medium and fine steady meshes assess absolute-load sensitivity. A short SA remeshing comparison on the medium mesh uses the same displacement statistics and amplitude limit but an independent nodal deformation field, combining resolution and disturbance changes.

Computational assistance Generative artificial intelligence (AI) tools, ChatGPT (OpenAI) and Claude (Anthropic), assisted the design of verification tests, code development, and numerical post-processing. The tabulated values and

data-driven figures are obtained from numerical outputs through the definitions given above, not from generative model predictions. *[Authors: specify the model versions and verify the scope of research assistance and the independent checks performed.]*

5. Results

5.1. Interpolation order verification

The smooth-field tests recover the nominal orders of the NN, BAR, and GP transfers, while CV2 depends on element geometry (Figure 1). NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 slopes slightly exceed three over this range, approaching approximately 3.2 on the finest pair, consistent with their nominal third-order construction. On the finest mixed grid, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3. CV2 is second order on the structured grid, with a fitted slope of 2.03, but reaches only 1.49 on the mixed grid and 1.35 on its finest pair, consistent with the reduced attainable order on quadrilaterals that remain distorted under refinement (Sec. 2.4). It returns the lumped density integral to within 4×10^{-15} on both grids, whereas the GP2 defect on the finest grids is 4.3×10^{-12} (structured) and 2.5×10^{-7} (mixed).

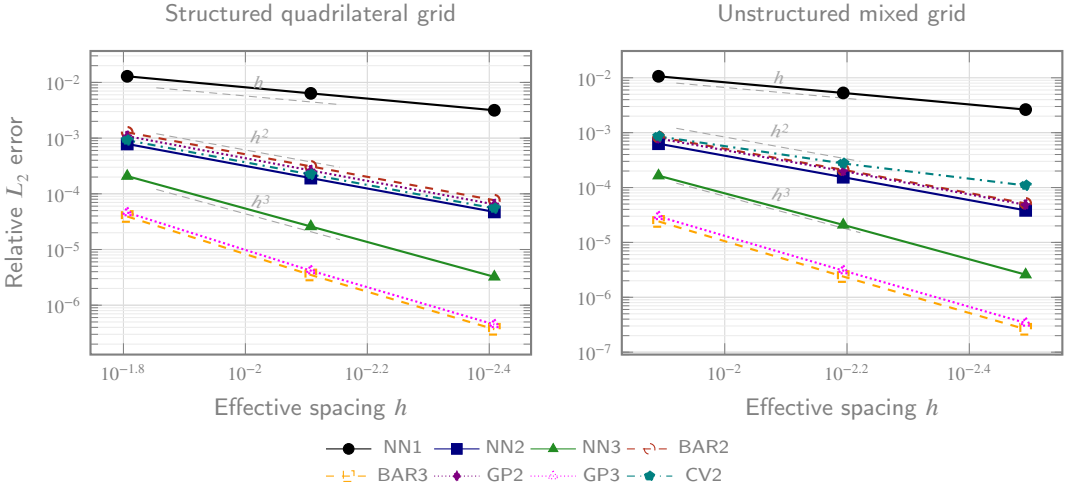


Figure 1: Smooth-field transfer accuracy on structured quadrilateral (left) and unstructured mixed (right) target meshes. The ordinate is the relative nodal density L_2 error; h is the count-based effective spacing. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer, with numerical suffixes indicating nominal accuracy order. Gray lines are first-, second-, and third-order slope guides, not fitted curves.

5.2. Boundedness and conservation

The circular-inclusion test shows that integral preservation does not ensure a bounded or admissible field (Table 2, structured transfer 64 → 128). Unlimited CV2 preserves the density integral to roundoff while exceeding the source bounds. Unlimited NN2 produces nonpositive thermal energy or pressure near the interface and is rejected as inadmissible. These failures concern different properties of the returned state.

Limiting suppresses the measured new extrema, but can change integral behavior. Where Barth–Jespersen limiting removes the NN2 reconstruction, its integral defect rises to the NN1 value. Adding the bounded global correction restores the integral without introducing density extrema. Hard-limited GP2 and CV2 also satisfy both density bounds and near-roundoff integral preservation in this test. Vertex-restricted GP3 remains bounded but retains a nonzero solver-integral defect, consistent with the representation difference in Appendix A.

For a separate smooth transfer, global correction reduces the density-integral defect to order 10^{-15} while changing the density L_1 error by at most 0.8%. The correction has a small one-transfer accuracy penalty in this test.

Table 2
Representative discontinuous-field transfers. Overshoot and undershoot are excursions beyond the exact density range $[1.2, 3.6]$; the last column is the source-to-target defect in the solver's lumped density integral. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction. Unlimited NN2 is not listed because its returned state is inadmissible.

Transfer	Overshoot	Undershoot	Integral defect
NN3, unlimited	0.4259	0.2251	-5.52×10^{-7}
NN2, BJ	0	0	-4.46×10^{-3}
NN2, BJ+GC	0	0	2.99×10^{-14}
BAR3, unlimited	0.1766	0.1811	-8.80×10^{-7}
BAR3, BJ+GC	0	0	5.34×10^{-14}
GP2, unlimited	0.1346	0.1357	-2.52×10^{-15}
GP2, hard limited	0	0	1.06×10^{-15}
GP3, hard limited, vertex output	0	0	-4.83×10^{-7}
CV2, unlimited	0.0730	0.1807	5.83×10^{-15}
CV2, hard limited	0	0	6.62×10^{-15}

5.3. Accumulated broadening under repeated Euler remeshing

Frequent mesh replacement broadens the density transition much more than it shifts its position (Table 3). With replacement every ten physical steps, every method locates the transition within 0.14 reference-grid cell of the fixed-mesh reference. Width changes are also small: 0.57 cell for NN1, 0.25 for BAR2, and at most 0.01 in magnitude for the remaining variants. At the higher frequency and $s = 4$, NN1 broadening grows to 21.41 cells and BAR2 broadening to 3.24 cells; GP2 has no width change to the reported precision, and CV2 changes the width by 0.01 cell. At $s = 2$, NN1 and BAR2 broaden by 26.36 and 22.91 cells, compared with 0.06 for both GP2 and CV2.

Table 3
Broadening of the density transition relative to the fixed-mesh reference. Width increases and the largest absolute position error among the listed methods are expressed in cells of the $h = 0.01$ reference mesh. GC is the global correction; $s = 2$ permits twice the displacement amplitude of $s = 4$.

Condition	Width increase					max.
	NN1	NN1+GC	BAR2	GP2	CV2	$\lvert\Delta x\rvert/h$
80 transfers, $s = 4$	0.57	–	0.25	0.00	0.00	0.14
398 transfers, $s = 4$	21.41	18.43	3.24	0.00	0.01	0.15
398 transfers, $s = 2$	26.36	25.65	22.91	0.06	0.06	0.91

The NN1, BAR2, and GP2 width ordering repeats in each deformation realization. At $s = 4$, excess widths range from 21.41 to 22.89 cells for NN1, 3.12 to 3.24 for BAR2, and 0 to 0.01 for GP2. At $s = 2$ the respective ranges are 24.98–27.14, 22.90–22.98, and 0.03–0.06 cells. CV2 width changes lie between -0.01 and 0.01 cell at $s = 4$ and between 0.05 and 0.07 cell at $s = 2$. Position errors remain below 0.5 cell for these uncorrected comparisons, confirming broadening as the dominant degradation.

For corrected NN1, the absolute value of the signed cumulative density-integral defect is at most 1.0×10^{-13} across all realizations, yet excess width remains 18.43–21.28 cells at $s = 4$ and 23.80–26.33 cells at $s = 2$. Integral correction leaves the broadened profile in place. CV2 holds the cumulative defect to at most 7.9×10^{-14} in magnitude, whereas the GP2 values are of order 10^{-6}–10^{-5}. Integral and profile preservation are therefore separate properties: NN1+GC attains the first without the second, GP2 the second with a small but finite integral defect, and CV2 both. At $s = 4$, NN2, NN3, BAR3, GP2, GP3, and CV2 all retain the transition much more closely than NN1 (Appendix B). At the stronger deformation, GP2 and CV2 have the smallest excess widths among the tested NN1, NN1+GC, BAR2, GP2, and CV2 variants.

5.4. Response to the two BDF2 histories

An older-history error alone produces a finite next-step lift response while the latest BDF2 history remains exact. In every transfer shown in Figure 2, the older-only and both-histories responses have opposite signs. For GP2, these differences are $+2.453 \times 10^{-3}$ and -7.315×10^{-3}, respectively, and for CV2 $+2.439 \times 10^{-3}$ and -7.272×10^{-3}.

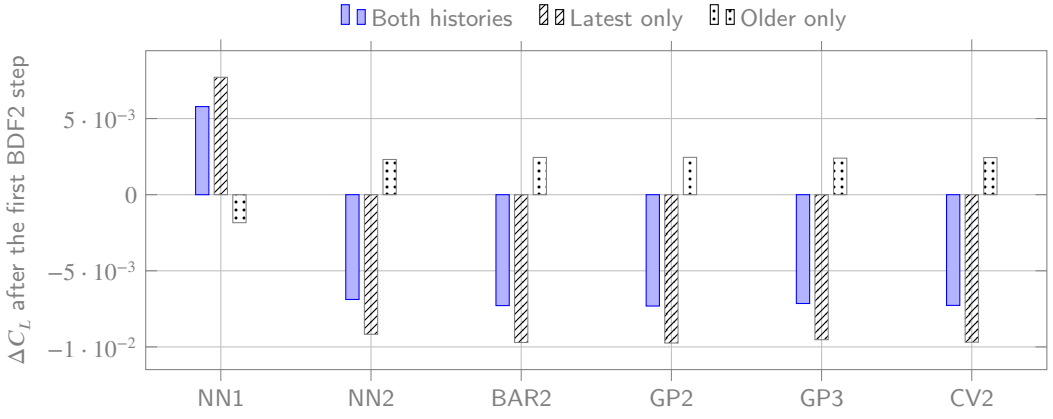


Figure 2: First-step lift-coefficient difference ΔC_L after transfer to the target Euler airfoil mesh, relative to its native continuation. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes indicate nominal accuracy order. Within each method, the same spatial discrepancy is assigned to both histories or to one history only. The opposite signs of the older-only and both-histories responses are consistent with the second-order backward differentiation weights.

At a common trial state, the measured residual differences agree with the exact BDF2 history-forcing identity to relative errors of approximately 10^{-7} for all tested transfers. Agreement also holds at the 370 slip-wall vertices, where the coincident source and target boundaries make the transferred histories compatible with the receiving slip condition. The separate wall-normal and tangential comparisons are reported in Appendix C.

Global correction barely changes the first-step lift in the NN1, BAR2, and CV2 controlled restarts. The NN1 both-histories response changes from 5.7873×10^{-3} to 5.7961×10^{-3}, BAR2 from -7.2852×10^{-3} to -7.2849×10^{-3}, and CV2, which already returns the source density integral, from -7.2724×10^{-3} to -7.2765×10^{-3}. These small changes are consistent with the spatial weighting of the load response in Eq. (25).

The target-referenced state norms combine transfer discrepancies with differences between native discrete equilibria. The tested second- and third-order transfers cluster near 7.0×10^{-4}, with less than 3% spread, whereas NN1 gives 1.47×10^{-3}. The small spread consequently describes restart discrepancies, not a ranking of the higher-order interpolation errors.

Across all eight transfer variants, the latest/full state-norm ratio is 1.3330–1.3333 and the older/full magnitude is 0.3335–0.3341. Volume-weighted state superposition defects of 1.8–3.5% and lift defects of 0.28–1.49% bound this directional check, with norm-dependent cancellation and amplitude results detailed in Appendix C. The ratios therefore support the predicted signed, directional response (Eq. (24)). They do not establish uniform linearity of the complete state response: the cancellation and amplitude tests in Appendix C retain unexplained volume-weighted discrepancies.

5.5. Accumulation and delayed Euler load response

GP2 and CV2 give smaller initial Euler drag trends than NN2 in both deformation realizations. Their slopes are 6.31×10^{-6} and 5.89×10^{-6} per 100 physical steps for GP2 and 5.92×10^{-6} and 5.82×10^{-6} for CV2, compared with 3.95×10^{-5} and 3.84×10^{-5} for NN2. The two-realization NN2 mean is approximately 6.4 times the GP2 mean and 6.6 times the CV2 mean. These fitted slopes characterize initial accumulation over the 400-step window, about 0.047 convective and 0.47 acoustic chord times; the smaller slopes describe this initial Euler response. Lift trends are less consistent between realizations.

A localized mesh change also produces a delayed pressure response beyond its immediate transfer footprint. This separates the initial interpolation disturbance from its subsequent evolution toward the airfoil. On the longer $0.20c$-band case, pressure-shell propagation speeds inferred from 10–75% amplitude thresholds range from 1.09 to 1.27 with GP2 and from 1.10 to 1.30 with CV2, bracketing $a_\infty = 1.183$; the spread reflects the detection thresholds for each single

event. Separately, the NN1 lift difference peaks three steps after the first post-transfer sample, at 4.66 times its initial value. The sampled second- and third-order responses peak at the first step and relax thereafter.

5.6. RANS transfer integrals and aerodynamic resolution

The RANS calculations distinguish two questions: how closely each transfer retains the monitored inventories, and whether those differences produce a resolved change in aerodynamic loads. GP2 and CV2 leave smaller monitored integral defects than NN2 (Table 4). The maximum GP2 density and total-energy defects are roughly two orders of magnitude below the NN2 values. Ratios of the separately maximized turbulence-inventory defects are approximately 247 for SA, and 52 and 37 for the SST k and ω inventories. CV2 holds every monitored inventory near roundoff, with mean-flow defects of at most 1.5×10^{-14} and turbulence-inventory defects of at most 4.6×10^{-12}. These instantaneous transfer diagnostics use the processing stages defined in Sec. 4.4, before the final SST positivity floor.

Table 4
Largest absolute source-relative integral change across the repeated RANS history transfers. Each column is maximized separately over the methods' two realizations. The SST values precede the final positivity floor. The inventory column identifies the integrand; the last column is the NN2/GP2 ratio of the separate maxima.

Closure	Inventory	NN2	GP2	CV2	NN2/GP2
SA	ρ	$8.41 \cdot 10^{-11}$	$4.56 \cdot 10^{-13}$	$7.19 \cdot 10^{-15}$	185
SA	ρu	$3.78 \cdot 10^{-9}$	$5.76 \cdot 10^{-11}$	$9.62 \cdot 10^{-15}$	66
SA	ρv	$1.65 \cdot 10^{-8}$	$8.08 \cdot 10^{-11}$	$1.08 \cdot 10^{-14}$	204
SA	ρE	$9.54 \cdot 10^{-11}$	$5.82 \cdot 10^{-13}$	$1.19 \cdot 10^{-14}$	164
SA	$\rho\tilde{\nu}$	$1.7 \cdot 10^{-6}$	$6.88 \cdot 10^{-9}$	$1.99 \cdot 10^{-12}$	247
SST	ρ	$8.25 \cdot 10^{-11}$	$4.67 \cdot 10^{-13}$	$1.44 \cdot 10^{-14}$	177
SST	ρu	$3.65 \cdot 10^{-9}$	$2.98 \cdot 10^{-11}$	$8.31 \cdot 10^{-15}$	123
SST	ρv	$1.64 \cdot 10^{-8}$	$7.77 \cdot 10^{-11}$	$9.37 \cdot 10^{-15}$	211
SST	ρE	$9.37 \cdot 10^{-11}$	$6.74 \cdot 10^{-13}$	$1.49 \cdot 10^{-14}$	139
SST	ρk	$2.22 \cdot 10^{-6}$	$4.23 \cdot 10^{-8}$	$4.61 \cdot 10^{-12}$	52
SST	$\rho\omega$	$7.98 \cdot 10^{-8}$	$2.15 \cdot 10^{-9}$	$9.29 \cdot 10^{-15}$	37

Over the repeated-remeshing interval, total-drag RMS deviations from the no-remeshing references range from 6.09×10^{-5} to 6.79×10^{-5} for the three methods. Paired NN2–GP2 differences have RMS 0.77×10^{-5} to 1.11×10^{-5}; the NN2–CV2 and GP2–CV2 values are 0.66–1.15×10^{-5} and 0.52–0.75×10^{-5}. The ordering of the three methods changes with closure and realization. Thus the similar overall response magnitudes conceal smaller differences between methods, whose temporal resolution determines whether an aerodynamic ordering is meaningful.

Time-step halving alters the paired difference substantially (Table 5). Across the three methods, the time-aligned individual lift histories change by 15.1–16.5% of their nominal-setting RMS for SA and 18.8–20.5% for SST; drag changes are 8.0–10.2% and 9.6–10.8%, respectively. Their RMS amplitudes change much less. The paired NN2–GP2 lift and drag differences change by 75.2% and 128.7% for SA, and by 146.1% and 110.3% for SST, with still larger changes in the SA quarter-step comparison. Comparable overall response magnitudes coexist with an unresolved detailed transient and method ranking.

Table 5
Time-step sensitivity at common physical times over the three-replacement RANS window. Values are $100S$ from Eq. (32), measuring RMS history differences. Individual columns give the largest NN2, GP2, or CV2 sensitivity; paired columns refer to the NN2–GP2 difference.

		Individual histories (%)		Paired difference (%)	
Closure	Step	C_L	C_D	C_L	C_D
SA	$\Delta t_0/2$	16.5	10.2	75.2	128.7
SA	$\Delta t_0/4$	18.8	15.0	131.5	171.0
SST	$\Delta t_0/2$	20.5	10.8	146.1	110.3

SA reaches the density-residual target at every reduced-step sample. Increasing the inner-iteration limit at the nominal step has a smaller, nonzero effect on the paired difference. For SST, most moved-grid samples still reach the increased iteration limit, and the $\Delta t_0/2$ calculation is incompletely converged. Its time-step sensitivity may therefore include iterative error. Appendix D reports these checks and the more sensitive friction contribution.

The steady viscous meshes show absolute-load sensitivity without establishing an asymptotic refinement sequence or bounding the paired NN2–GP2 error on a common mesh. The medium-mesh transient also changes the nodal disturbance, so refinement alone cannot explain its response. The evidence consequently establishes smaller monitored transfer-integral defects for GP2 and CV2, but no resolved aerodynamic advantage over NN2. In particular, the large sensitivity of the paired NN2–GP2 response precludes using the nominal histories to infer a three-method load ranking.

5.7. Spatial fidelity and restart response

The comparisons identify two distinct routes by which transfer changes subsequent evolution. Repeated remeshing modifies spatial structure: NN1+GC retains the density total but leaves a broadened transition, whereas GP2 and CV2 retain the profile much more closely. The controlled restarts isolate the temporal route: the receiving BDF2 equation combines discrepancies from both histories, so an exact latest state does not remove an older-history disturbance. These observations explain why a small integral defect alone cannot characterize restart accuracy.

The distinction between transfer discrepancy and native mesh sensitivity is equally important. The similar target-referenced discrepancies of higher-order methods in the controlled airfoil restarts include differences between the two native discrete equilibria and therefore do not isolate interpolation error. This does not contradict their different smooth-field approximation errors. Likewise, smaller integral defects in RANS do not necessarily produce smaller load errors: the integrals constrain selected global quantities, whereas the load response depends on the spatial structure of the discrepancy and the receiving dynamics. This interpretation is consistent with constrained-transfer studies that distinguish integral preservation from other discrete properties [6], and with balance-preserving transfer designed to suppress subsequent numerical disturbances [12, 13].

CV2 combines near-roundoff global integral preservation with small transition broadening in the repeated Euler tests. Its reduced order on distorted quadrilaterals exposes a trade-off between the chosen integral-compatible representation and affine reproduction. Limiting and boundary treatment further distinguish its raw local constraints from the returned state. Across the experiments, the useful comparison is consequently between the properties retained by each transfer and their measured consequences for the evolving solution; in RANS, those consequences remain numerically resolved at the transfer level but unresolved in the paired load response.

6. Conclusions

Mesh replacement in a multistep calculation perturbs both the spatial representation and the histories used to advance the solution. The controlled BDF2 restarts show that an older-history discrepancy produces a finite next-step lift response even when the latest history is exact. Its sign is opposite to the response to the same discrepancy in the latest history, consistent with the BDF2 weights. The exact residual identity explains this forcing; the computed state and load responses remain dependent on the receiving dynamics and the accuracy of the physical-step solve.

Integral preservation and spatial fidelity have different consequences under repeated transfer. Globally corrected NN1 retains the signed cumulative density integral to roundoff while broadening the transition by 18–26 reference cells. CV2 retains the same integral with at most 0.07 cell of broadening, comparable to GP2 in profile preservation. Correcting a global total therefore cannot recover spatial structure lost during transfer. Boundedness is independent of both properties, as the overshoots of unlimited CV2 demonstrate.

The fixed-body RANS calculations extend this distinction to turbulence transport and viscous loads. GP2 reduces the monitored turbulence-inventory defects relative to NN2, and CV2 keeps the monitored defects near roundoff at the stated processing stages. These differences do not establish an aerodynamic ranking: paired load differences remain sensitive to time-step refinement, with incomplete inner convergence adding uncertainty for SST. The smaller initial Euler drag trends of GP2 and CV2 are thus a finding for the tested Euler configurations, rather than evidence of a general RANS advantage.

The study shows that a faithful restart depends on the spatial distribution and temporal placement of transfer discrepancies, as well as the quantities preserved globally. Conservation is valuable, but its effect on subsequent flow evolution cannot be inferred independently of the representation, the transferred histories, and the numerical resolution of the receiving calculation.

Appendix A. Representation and reference-state identities

GP3 reconstructs the source value at the midpoint of edge (i, j) as

$$\phi_{ij} = \tfrac{1}{2}(\phi_i + \phi_j) - \tfrac{1}{8} d_{ij}^T \overline{H}_{ij} d_{ij}, \qquad d_{ij} = x_i - x_j, \qquad \overline{H}_{ij} = \tfrac{1}{2}(H_i + H_j). \tag{A.1}$$

For a quadratic field, the curvature term equals the difference between the endpoint average and midpoint value. The complete quadratic projection can thus be exact while vertex restriction changes the solver integral. For example, $f(x) = x^2$ on a uniform partition of $[0, 1]$ belongs to the quadratic finite-element space, but integrating only its vertex values with linear nodal weights gives

$$Q_h(f) = \frac{1}{3} + \frac{h^2}{6}. \tag{A.2}$$

This second-order discrepancy comes entirely from vertex restriction; the complete projection remains exact.

Appendix A.1. Discrepancy relative to a native target state

Let $\Pi_M u^k$ represent a common exact field in the solver representation, with native error $e_M^k = U_M^k - \Pi_M u^k$. For the complete, possibly nonlinear, transfer map,

$$\begin{aligned} \varepsilon^k = & \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}} \\ & + \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{source and target discretization contributions}} . \end{aligned} \tag{A.3}$$

Only a linear map permits the second line to be written $\mathcal{I}_{AB} e_A^k - e_B^k$. A load difference likewise separates as

$$\begin{aligned} G_B(\widehat{U}_B^k) - G_A(U_A^k) = & \, G_B(U_B^k) - G_A(U_A^k) \\ & + G_B(\widehat{U}_B^k) - G_B(U_B^k). \end{aligned} \tag{A.4}$$

The first difference is native mesh sensitivity; the second is the target-referenced restart response. Neither alone is an independently known continuum error.

Appendix B. Repeated Euler transfer comparisons

Table B.1 includes the higher-order alternatives to the main-text comparison, and Table B.2 reports all three realizations, including globally corrected NN1. Cancellation of signed transfer-integral changes leaves local distortion unconstrained, so the tables report integral and profile errors separately.

Table B.1
Frequency comparison at safety factor $s = 4$ for all transfer variants. Δn is the number of physical steps between mesh changes; $\Delta x/h$ and $\Delta w/h$ are signed position and width changes relative to the fixed-mesh reference.

Transfer	Δn	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho$
NN1	10	0.14	0.57	$1.1 \cdot 10^{-4}$
NN1	2	0.1	21.41	$5 \cdot 10^{-4}$
NN2	10	−0.02	−0.01	$-8.8 \cdot 10^{-6}$
NN2	2	−0.04	0	$-1.7 \cdot 10^{-5}$
NN3	10	0.01	−0.01	$2.2 \cdot 10^{-5}$
NN3	2	0.02	0	$2.7 \cdot 10^{-5}$
BAR2	10	0.01	0.25	$2 \cdot 10^{-5}$
BAR2	2	−0.03	3.24	$1.4 \cdot 10^{-4}$
BAR3	10	0	0.01	$-1.7 \cdot 10^{-5}$
BAR3	2	0	0.07	$-5.2 \cdot 10^{-6}$
GP2	10	0.01	0	$2.6 \cdot 10^{-6}$
GP2	2	0.01	0	$-2.8 \cdot 10^{-6}$
GP3	10	−0.01	0	$-2.1 \cdot 10^{-5}$
GP3	2	−0.01	0.02	$1.8 \cdot 10^{-5}$
CV2	10	0	0	$3 \cdot 10^{-14}$
CV2	2	0	0.01	$-8.2 \cdot 10^{-15}$

Table B.2
High-frequency Euler comparison across three random deformation realizations. GC indicates whether the global correction is applied; $s = 2$ permits twice the displacement amplitude of $s = 4$. Position and width changes are in cells of the $h = 0.01$ reference mesh. The final column is the signed sum of target-minus-source density-integral changes at transfer.

Transfer	GC	s	Realization	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho$
BAR2	No	2	1	−0.06	22.91	$2.7 \cdot 10^{-4}$
BAR2	No	2	2	−0.03	22.98	$3.1 \cdot 10^{-4}$
BAR2	No	2	3	−0.07	22.9	$1.9 \cdot 10^{-4}$
BAR2	No	4	1	−0.03	3.24	$1.4 \cdot 10^{-4}$
BAR2	No	4	2	0	3.2	$1.7 \cdot 10^{-4}$
BAR2	No	4	3	−0.04	3.12	$6.6 \cdot 10^{-5}$
CV2	No	2	1	0.19	0.06	$-9.5 \cdot 10^{-15}$
CV2	No	2	2	0.2	0.07	$-4.2 \cdot 10^{-14}$
CV2	No	2	3	0.18	0.05	$7.9 \cdot 10^{-14}$
CV2	No	4	1	0	0.01	$-8.2 \cdot 10^{-15}$
CV2	No	4	2	0.01	−0.01	$6.8 \cdot 10^{-14}$
CV2	No	4	3	0	0.01	$4.6 \cdot 10^{-14}$
GP2	No	2	1	0.17	0.06	$-1 \cdot 10^{-5}$
GP2	No	2	2	0.19	0.06	$-9.6 \cdot 10^{-6}$
GP2	No	2	3	0.19	0.03	$-9.5 \cdot 10^{-6}$
GP2	No	4	1	0.01	0	$-2.8 \cdot 10^{-6}$
GP2	No	4	2	0.02	0	$-3.1 \cdot 10^{-6}$
GP2	No	4	3	0.01	0.01	$-2.5 \cdot 10^{-6}$
NN1	Yes	2	1	−0.91	25.65	$-2.8 \cdot 10^{-15}$
NN1	Yes	2	2	−0.65	23.8	$7.9 \cdot 10^{-14}$
NN1	Yes	2	3	0.01	26.33	$1 \cdot 10^{-13}$
NN1	Yes	4	1	−0.15	18.43	$-5.3 \cdot 10^{-14}$
NN1	Yes	4	2	−0.01	21.28	$9.5 \cdot 10^{-15}$
NN1	Yes	4	3	−0.11	21.2	$-1.9 \cdot 10^{-14}$
NN1	No	2	1	−0.38	26.36	$1.3 \cdot 10^{-3}$
NN1	No	2	2	−0.29	24.98	$-1.6 \cdot 10^{-3}$
NN1	No	2	3	0.44	27.14	$4.9 \cdot 10^{-4}$
NN1	No	4	1	0.1	21.41	$5 \cdot 10^{-4}$
NN1	No	4	2	0.3	22.89	$-8.9 \cdot 10^{-4}$
NN1	No	4	3	0.2	22.75	$8.3 \cdot 10^{-4}$

Appendix C. Controlled-history diagnostics

Appendix C.1. Directional response and boundary enforcement

Table C.1 reports all response ratios and superposition defects. State ratios use the volume-weighted norm, while the older-only force ratio retains its sign. The defects quantify the departure from directional superposition alongside the coefficient comparison.

Table C.1
Euler BDF2 response ratios and relative superposition defects. Subscripts f, n, and $n-1$ denote both-histories, latest-only, and older-only perturbations. Ideal directional ratios are $4/3$ and $-1/3$; the state columns show their nonnegative norm ratios.

Transfer	e^U_{sup}	$\|\delta_n\|_M/\|\delta_f\|_M$	$\|\delta_{n-1}\|_M/\|\delta_f\|_M$	$e^{C_L}_{\mathrm{sup}}$	$\Delta C_{L,n}/\Delta C_{L,f}$	$\Delta C_{L,n-1}/\Delta C_{L,f}$
NN1	$1.8\cdot10^{-2}$	1.3333	0.3335	$1.5\cdot10^{-2}$	1.3329	−0.3179
NN2	$3.5\cdot10^{-2}$	1.3331	0.3341	$4.2\cdot10^{-3}$	1.3318	−0.336
NN3	$3.4\cdot10^{-2}$	1.3331	0.334	$5.3\cdot10^{-3}$	1.3309	−0.3362
BAR2	$3.4\cdot10^{-2}$	1.3332	0.334	$3.1\cdot10^{-3}$	1.3328	−0.3358
BAR3	$3.3\cdot10^{-2}$	1.3332	0.334	$2.8\cdot10^{-3}$	1.3329	−0.3357
GP2	$3.4\cdot10^{-2}$	1.333	0.334	$3\cdot10^{-3}$	1.3323	−0.3353
GP3	$3.4\cdot10^{-2}$	1.3331	0.334	$3.1\cdot10^{-3}$	1.3327	−0.3358
CV2	$3.4\cdot10^{-2}$	1.333	0.334	$3.1\cdot10^{-3}$	1.3323	−0.3354

At a common trial state and the 370 slip-wall vertices, the wall-normal momentum mismatch is 3.4×10^{-12} for GP2, 4.1×10^{-12} for GP3 and CV2, and 4.5×10^{-12} for NN1. The tangential mismatch is $(1.4\text{–}1.7)\times10^{-11}$, a fraction $(1.1\text{–}1.3)\times10^{-7}$ of the tangential history forcing of 1.25×10^{-4}. All four transfers return the same wall state, that of the source solution; its wall-normal momentum discrepancy from the native target state is 1.1×10^{-14}, so the normal history forcing is negligible and the slip-wall operation of the receiving equations does not alter it. Comparing the measured residual with its predicted history forcing still requires the same boundary operation on both.

Appendix C.2. Cancellation of the initial history forcing

For an imposed target-history discrepancy e, setting $\varepsilon^n = e/4$ and $\varepsilon^{n-1} = e$ makes η^n zero. At a common trial state, the two BDF2 equations are identical; an exact solve on the same local solution branch should return the same next state. The measurements distinguish force agreement from state agreement (Table C.2). First-step lift differences are of order 10^{-9}, whereas relative state differences are approximately 2×10^{-4} in the Euclidean norm and 0.043–0.090 in the volume-weighted norm. Neither norm has an independently measured iterative floor, and the small lift difference leaves a finite state discrepancy.

Table C.2
Initial history-cancellation test. Both histories are non-native but $2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. First-step force differences and state differences use the same native continuation; both state norms are shown to avoid interpreting a small load difference as complete state agreement.

Transfer	ΔC_L	ΔC_D	$\|\delta\|_2/\|e\|_2$	$\|\delta\|_M/\|e\|_M$
CV2	$7.59\cdot10^{-9}$	$-5.54\cdot10^{-9}$	$2.43\cdot10^{-4}$	$8.98\cdot10^{-2}$
GP2	$7.64\cdot10^{-9}$	$-5.54\cdot10^{-9}$	$2.43\cdot10^{-4}$	$8.98\cdot10^{-2}$
NN1	$2.11\cdot10^{-9}$	$-4.1\cdot10^{-9}$	$2.1\cdot10^{-4}$	$4.34\cdot10^{-2}$

The initial residual probes in this additional test use different trial states. For a trial displacement q, their difference has the form

$$\Delta\Phi = \frac{3M_B q}{2\Delta t} + F_B(Y+q) - F_B(Y) - \frac{M_B\eta^n}{\Delta t}. \tag{C.1}$$

With $q = e/4$ and $\eta^n = 0$, the temporal part is three quarters of the older-only forcing magnitude. The measured ratio near 0.75 also includes a spatial-residual difference and therefore is not a common-trial check of Eq. (21). Cancellation is temporary even for an exact first step: after $\delta^{n+1} = 0$, the older level retains $e/4$, producing the subsequent history combination $-e/8$.

Appendix C.3. Perturbation-amplitude dependence

Scaling the discrepancy by $\lambda = 1, 1/2, 1/4$ reduces the Euclidean-state and lift superposition remainders for BAR2, GP2, and CV2 (Table C.3). The Euclidean absolute state remainder decreases by factors of about 3.2–3.5 per halving, faster than linearly over this range. The volume-weighted absolute remainder falls more slowly, from approximately 2.4×10^{-5} to 1.5–1.7×10^{-5}, while its normalized value rises from about 3.4% to 8.7–9.2%. The directional response is diagnostic- and norm-dependent; these tests establish neither a uniform quadratic remainder nor a verified numerical floor in all norms.

Table C.3
Amplitude dependence of first-step superposition. $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$ is the state remainder and $\mathcal{E}_L$ its lift counterpart. Relative remainders use the corresponding both-histories response. The Euclidean and volume-weighted state metrics have different amplitude dependence.

Transfer	λ	$\|\mathcal{E}\|_2$	$\|\mathcal{E}\|_2/\|\delta_f\|_2$	$\|\mathcal{E}\|_M$	$\|\mathcal{E}\|_M/\|\delta_f\|_M$	$\|\mathcal{E}_L\|$
BAR2	1	$1.76 \cdot 10^{-3}$	$7.78 \cdot 10^{-3}$	$2.36 \cdot 10^{-5}$	$3.36 \cdot 10^{-2}$	$2.22 \cdot 10^{-5}$
BAR2	0.5	$5.45 \cdot 10^{-4}$	$4.82 \cdot 10^{-3}$	$1.97 \cdot 10^{-5}$	$5.6 \cdot 10^{-2}$	$4.74 \cdot 10^{-6}$
BAR2	0.25	$1.6 \cdot 10^{-4}$	$2.83 \cdot 10^{-3}$	$1.54 \cdot 10^{-5}$	$8.74 \cdot 10^{-2}$	$1.76 \cdot 10^{-6}$
GP2	1	$1.86 \cdot 10^{-3}$	$8.23 \cdot 10^{-3}$	$2.45 \cdot 10^{-5}$	$3.44 \cdot 10^{-2}$	$2.23 \cdot 10^{-5}$
GP2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.42 \cdot 10^{-6}$
GP2	0.25	$1.65 \cdot 10^{-4}$	$2.93 \cdot 10^{-3}$	$1.65 \cdot 10^{-5}$	$9.24 \cdot 10^{-2}$	$1.58 \cdot 10^{-6}$
CV2	1	$1.86 \cdot 10^{-3}$	$8.22 \cdot 10^{-3}$	$2.42 \cdot 10^{-5}$	$3.4 \cdot 10^{-2}$	$2.27 \cdot 10^{-5}$
CV2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.53 \cdot 10^{-6}$
CV2	0.25	$1.66 \cdot 10^{-4}$	$2.95 \cdot 10^{-3}$	$1.64 \cdot 10^{-5}$	$9.16 \cdot 10^{-2}$	$1.63 \cdot 10^{-6}$

Appendix D. RANS numerical sensitivity

Appendix D.1. Inner iterations and time-step refinement

Table D.1 separates achieved residual criteria from iteration limits. Nominal-setting rows cover the full repeated-remeshing histories; other rows cover the short sensitivity calculations. SA reaches the density criterion throughout the 1600-iteration sensitivity window and at every half- and quarter-step point. SST reaches the iteration limit on 28–30 of 38 steps even with 6400 inner iterations, and on 56–60 of 78 half-step points. These SST references remain incompletely converged.

Table D.1
Inner-iteration behavior for the paired calculations of the first deformation realization. Entries separated by slashes are GP2/NN2/CV2. $N_{\max}$ is the iteration limit; residuals are median terminal $\log_{10}\mathrm{rms}(r_\rho)$. Nominal-setting rows cover the full interval; the remaining rows cover the short numerical-sensitivity window.

Closure	$N_{\max}$	$\Delta t/\Delta t_0$	Physical steps	Steps at limit	Residual
SA	800	1	398	152/155/147	-12.00/-12.00/-12.00
SA	1600	1	38	0/0/0	-12.00/-12.00/-12.00
SA	800	1/2	78	0/0/0	-12.00/-12.00/-12.00
SA	800	1/4	158	0/0/0	-12.00/-12.00/-12.00
SST	800	1	398	333/324/336	-11.10/-11.24/-11.11
SST	1600	1	38	28/28/30	-11.33/-11.33/-11.27
SST	6400	1	38	28/30/29	-11.27/-11.17/-11.18
SST	800	1/2	78	56/58/60	-11.27/-11.32/-11.23

Increasing the iteration limit changes the SA lift and total-drag histories by at most 0.3% of their own RMS and their smaller paired differences by 8.4–10.9% (Table D.2, Eq. (32)). For SST, the 6400-cap comparison changes the paired lift difference by 67.7% of its nominal-setting RMS. Residual stagnation accompanies this sensitivity without identifying a unique cause in the turbulence equations.

In the SA step-refinement sequence, the paired lift RMS is approximately 1.21×10^{-5}, 1.60×10^{-5}, and 2.08×10^{-5}; the drag values are 2.19×10^{-6}, 3.77×10^{-6}, and 4.55×10^{-6}. Although individual RMS amplitudes remain closer across the same samples, the detailed paired histories are unresolved. Friction drag is more sensitive than total drag:

Table D.2
Effect of increasing the inner-iteration limit, relative to the nominal setting. Columns have the same definition as Table 5: individual values are the largest NN2, GP2, or CV2 sensitivity, and paired values refer to the NN2–GP2 difference. All entries are percentages.

		Individual histories		Paired difference	
Closure	Iteration limit	C_L	C_D	C_L	C_D
SA	1600	0.3	0.2	10.9	8.4
SST	1600	4.5	1.4	13.7	6.0
SST	6400	8.5	2.6	67.7	25.5

the SA quarter-step history differences are approximately 76–92% of the nominal-setting friction-response RMS. The smaller total-drag sensitivity consequently leaves componentwise convergence unestablished.

Appendix D.2. Steady mesh sensitivity and cross-mesh robustness

The steady meshes in Table D.3 preserve the first off-wall spacing while changing the remaining resolution. Lift, total drag, and pressure drag decrease monotonically; friction drag varies nonmonotonically. For this anisotropic family, the count-based spacing describes mesh density without establishing a common asymptotic error expansion. The comparison reports absolute-load sensitivity, with no extrapolated loads or discretization-uncertainty estimate; it does not quantify the error in the paired transfer difference on a common mesh.

Table D.3
Steady aerodynamic loads without remeshing on the coarse, medium, and fine viscous meshes. The count-based descriptor $h/h_c = \sqrt{N_c/N}$ uses the coarse-mesh vertex count N_c as reference; it is not a uniform refinement ratio in every direction. The same first off-wall spacing is used on all levels.

Closure	Mesh	Vertices	h/h_c	C_L	C_D	$C_{D,p}$	$C_{D,v}$
SA	Coarse	23,271	1	0.67626	0.010229	0.003559	0.006671
SA	Medium	32,102	0.851	0.67084	0.010048	0.003401	0.006647
SA	Fine	48,228	0.695	0.66854	0.009996	0.00333	0.006665
SST	Coarse	23,271	1	0.67561	0.009858	0.003469	0.006389
SST	Medium	32,102	0.851	0.6686	0.009705	0.003321	0.006385
SST	Fine	48,228	0.695	0.66639	0.009644	0.003247	0.006397

On the medium mesh, the short SA remeshing interval gives individual lift RMS responses near 1.07×10^{-4} and drag responses near 5.39×10^{-5}; the paired differences are 8.77×10^{-6} and 9.94×10^{-7}. The methods remain relatively close, but the new node positions also change the random displacement field. Resolution alone cannot explain the smaller response. Five to seven of the 38 physical steps reach the nominal inner-iteration limit.

Data availability

The numerical tables supporting the figures and comparisons, together with the post-processing routines needed to reproduce the reported reductions, are available in the public data repository. The archive does not include every flow-field output from the simulations.

Funding

[Authors: supply funders and grant identifiers, or confirm that there was no specific funding.]

Contributor Roles Taxonomy (CRediT) authorship contribution statement

[Authors: assign CRediT roles to each named author.]

Declaration of competing interest

[Authors: supply the declaration of financial and non-financial competing interests.]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) and Claude (Anthropic) were used to assist literature synthesis, organization, scientific editing, and the preparation of the manuscript and data-driven figures. *[Authors: confirm the tools and versions used, describe human review and verification, and provide the final statement accepting responsibility for the published content.]*

References

[1] F. Alauzet, A. Loseille, G. Olivier, Time-accurate multi-scale anisotropic mesh adaptation for unsteady flows in CFD, J. Comput. Phys. 373 (2018) 28–63. doi:10.1016/j.jcp.2018.06.043.

[2] B. Sauvage, F. Alauzet, A. Dervieux, A space and time fixed point mesh adaptation method, J. Comput. Phys. 519 (2024) 113389. doi:10.1016/j.jcp.2024.113389.

[3] T. Levý, G. May, Towards a robust time-accurate anisotropically adaptive hybridized discontinuous Galerkin method, Comput. Fluids 301 (2025) 106792. doi:10.1016/j.compfluid.2025.106792.

[4] X. Jiao, M. T. Heath, Common-refinement-based data transfer between non-matching meshes in multiphysics simulations, Int. J. Numer. Methods Eng. 61 (2004) 2402–2427.

[5] P. E. Farrell, J. R. Maddison, Conservative interpolation between volume meshes by local Galerkin projection, Comput. Methods Appl. Mech. Eng. 200 (1–4) (2011) 89–100. doi:10.1016/j.cma.2010.07.015.

[6] J. R. Maddison, H. R. Hiester, Optimal constrained interpolation in mesh-adaptive finite element modeling, SIAM J. Sci. Comput. 39 (5) (2017) A2257–A2286. doi:10.1137/15M102054X.

[7] J. K. Dukowicz, J. W. Kodis, Accurate conservative remapping (rezoning) for arbitrary Lagrangian–Eulerian computations, SIAM J. Sci. Stat. Comput. 8 (3) (1987) 305–321. doi:10.1137/0908037.

[8] L. G. Margolin, M. Shashkov, Second-order sign-preserving conservative interpolation (remapping) on general grids, J. Comput. Phys. 184 (1) (2003) 266–298. doi:10.1016/S0021-9991(02)00033-5.

[9] F. Alauzet, M. Mehrenberger, P^1-conservative solution interpolation on unstructured triangular meshes, Int. J. Numer. Methods Eng. 84 (13) (2010) 1552–1588. doi:10.1002/nme.2951.

[10] F. Alauzet, A parallel matrix-free conservative solution interpolation on unstructured tetrahedral meshes, Comput. Methods Appl. Mech. Eng. 299 (2016) 116–142. doi:10.1016/j.cma.2015.10.012.

[11] T. Levý, G. May, Conservative solution transfer between anisotropic meshes for time-accurate hybridized discontinuous Galerkin methods, Int. J. Numer. Methods Fluids 96 (6) (2024) 1011–1030. doi:10.1002/fld.5278.

[12] J. R. Maddison, C. J. Cotter, P. E. Farrell, Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling, Ocean Model. 37 (1–2) (2011) 35–48. doi:10.1016/j.ocemod.2010.12.007.

[13] J. R. Maddison, C. J. Cotter, P. E. Farrell, Corrigendum to "geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling" [Ocean Model. 37 (2011) 35–48], Ocean Model. 68 (2013) 106. doi:10.1016/j.ocemod.2013.04.007.

[14] T. J. Barth, P. O. Frederickson, Higher order solution of the Euler equations on unstructured grids using quadratic reconstruction, in: 28th Aerospace Sciences Meeting, AIAA Paper 90-0013, Reno, Nevada, 1990.

[15] T. J. Barth, Recent developments in high order k-exact reconstruction on unstructured meshes, in: 31st Aerospace Sciences Meeting, AIAA Paper 93-0668, Reno, Nevada, 1993.

[16] D. J. Mavriplis, Revisiting the least-squares procedure for gradient reconstruction on unstructured meshes, in: 16th AIAA Computational Fluid Dynamics Conference, Orlando, Florida, 2003, AIAA Paper 2003-3986. doi:10.2514/6.2003-3986.

[17] B. Diskin, J. L. Thomas, Accuracy of gradient reconstruction on grids with high aspect ratio, Tech. Rep. NIA Report 2008-12, National Institute of Aerospace, Hampton, Virginia (2008).

[18] T. J. Barth, D. C. Jespersen, The design and application of upwind schemes on unstructured meshes, in: 27th Aerospace Sciences Meeting, AIAA Paper 89-0366, Reno, Nevada, 1989. doi:10.2514/6.1989-366.

[19] D. A. Dunavant, High degree efficient symmetrical Gaussian quadrature rules for the triangle, Int. J. Numer. Methods Eng. 21 (6) (1985) 1129–1148.

[20] P. Keast, Moderate-degree tetrahedral quadrature formulas, Comput. Methods Appl. Mech. Eng. 55 (1986) 339–348.

[21] M. Shashkov, B. Wendroff, The repair paradigm and application to conservation laws, J. Comput. Phys. 198 (1) (2004) 265–277.

[22] M. Kucharik, M. Shashkov, B. Wendroff, An efficient linearity-and-bound-preserving remapping method, J. Comput. Phys. 188 (2) (2003) 462–471.

[23] B. Koobus, C. Farhat, Second-order time-accurate and geometrically conservative implicit schemes for flow computations on unstructured dynamic meshes, Comput. Methods Appl. Mech. Eng. 170 (1–2) (1999) 103–129. doi:10.1016/S0045-7825(98)00207-2.

[24] P. Geuzaine, C. Grandmont, C. Farhat, Design and analysis of ALE schemes with provable second-order time-accuracy for inviscid and viscous flow simulations, J. Comput. Phys. 191 (1) (2003) 206–227. doi:10.1016/S0021-9991(03)00311-5.

[25] P. R. Spalart, S. R. Allmaras, A one-equation turbulence model for aerodynamic flows, Rech. Aérosp. (1) (1994) 5–21.

[26] F. R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, AIAA J. 32 (8) (1994) 1598–1605. doi:10.2514/3.12149.

[27] T. D. Economon, F. Palacios, S. R. Copeland, T. W. Lukaczyk, J. J. Alonso, SU2: An open-source suite for multiphysics simulation and design, AIAA J. 54 (3) (2016) 828–846. doi:10.2514/1.J053813.